

DILSHAN PATHEGAMAGE

Trainee Software Engineer | Final Year IT Undergraduate

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PROFESSIONAL SUMMARY

Trainee Software Engineer and Final-year IT undergraduate with proven experience in developing scalable full-stack applications, implementing AI-driven solutions, and supporting DevOps workflows. Experienced in MERN stack, Flutter, and cloud-based deployments, with a strong focus on performance, clean architecture, and real-world problem-solving. Seeking a Software Engineer role to contribute effectively to high-impact projects while growing as an IT professional.

WORK EXPERIENCE

Trainee Software Engineer at Virstack Pvt Ltd

DEC 2025 - JUN 2026

- Leading and contributing to the development of scalable full-stack web and mobile applications using MERN stack and Flutter/Dart in production environments.
- Designing and optimizing RESTful APIs, real-time communication systems, and application architectures with a focus on performance, maintainability, and scalability.
- Collaborating on AI-integrated solutions, cloud deployments, and DevOps workflows using Docker, AWS Lightsail, Nginx, and CI/CD within agile international teams.

Intern Software Engineer at Virstack Pvt Ltd.

MAR 2025 - NOV 2025

- Developed and optimized full-stack web and mobile applications using MERN stack and Flutter/Dart, contributing to production-ready features and scalable backend services.
- Worked on AI workflows, including LLM fine-tuning, and implemented a Bluetooth GATT server enabling real-time VR-mobile communication.
- Supported deployment and infrastructure tasks using Docker, AWS Lightsail, Nginx, and CI/CD while collaborating in agile development environments.

EDUCATION

University of Moratuwa

Bsc.(Hons) in Information Technology

JUN 2022 - JUL 2026

- Level 2 Semester 2 SGPA : **3.90 - Dean's List**
- Current GPA - 3.51 (out of 4.0 - up to L3S2)

Galahitiyawa Central College - Ganemulla

G.C.E. Advanced Level

OCT 2020

- Combined Maths - A | Chemistry - B | Physics - B

TECHNICAL SKILLS

Programming Languages : JavaScript, TypeScript, Dart, Java, C#, Python

Web & Mobile Development : React Js, Node Js, Flutter, Asp.Net, Tailwind CSS, Bootstrap

Databases : Mongo DB, MYSQL, MSSQL, Neo4j

Cloud & DevOps : AWS Lightsail, AWS S3, AWS EC2, CI/CD Pipelines, Docker, NGINX

Testing & Quality Assurance : Cypress, Jest, Cucumber

PROJECTS

Final Year Research - Exploring Hardware-Software Codesign of Spatial Deep Learning Hardware Accelerators

(Group Research - Ongoing)

Research on developing a cycle-accurate hardware-software co-design simulator framework for systolic array-based DNN accelerators, addressing critical performance bottlenecks in DNN execution on computing systems. Implementing SystemVerilog RTL models coupled with a Python compiler/mapper, exploring loop optimizations, memory management schemes, and multi-DNN scheduling algorithms.

- Role - Researcher
- Technologies - Python, System Verilog, DNN Workloads, FPGA Simulation

Sri Lanka Tourist Arrivals Prediction Using Machine Learning (Individual Project)

Developed a machine learning-based forecasting system to predict Sri Lanka's monthly foreign tourist arrivals using tourism data from 2019-2024. Built and optimized a Random Forest regression model with advanced feature engineering, time-series forecasting, and SHAP explainability, achieving 92% R² prediction accuracy.

- Role - ML Engineer, Full Stack Developer
- Technologies - Python, Scikit-learn, SHAP, Machine Learning, Data Analysis

[GitHub repository](#)

Expert System - University Course Selection Advisor (Individual Project)

Developed a rule-based expert system to assist university course selection based on semester, prerequisites, credit requirements, and targeted career paths. Implemented career-based course recommendations, prerequisite validation, and automated credit validation using a Prolog knowledge base, with an interactive Python GUI for user-friendly decision support.

- Role - Full Stack Developer
- Technologies - Prolog, Python, Tkinter

[GitHub repository](#)

TicketMate - Public Transport E-Ticketing System (Level 2 Software Project - Mentored by Bistec Global)

The public transport-focused web application offers train and bus searches, seat reservations with online payments, reservation changes and cancellations with refunds, live tracking, emergency updates, vehicle/driver/vehicle owner/passenger registration, scheduling, and income prediction. I developed features for searches, reservations, changes and cancellations of reservations, passenger profile management, and a review and rating system.

- Role - Full Stack Developer, Ui/Ux Designer
- Technologies - React Js, TypeScript, Asp.Net, MsSql, Bootstrap, Azure Cloud

[Front-end repository](#)

[Back-end repository](#)

MozzaMelt - Ecommerce Platform (Group Project)

Built a full-featured pizza ordering web application with payment, order tracking, reviews, and admin management. Developed the review and rating service, including backend logic and unit testing.

- Role - Full Stack Developer
- Technologies - React Js, Asp.Net Core, MsSql, Docker, Ci/Cd Pipeline, Xunit

[Front-end repository](#)

[Back-end repository](#)

NeurA - Chatbot (Individual Project)

NeurA is a dynamic chatbot platform designed to provide quick, accurate, and well-organized responses integrated with the Google Gemini API.

- Role - Full Stack Developer, Ui/Ux Designer
- Technologies - React Js, TypeScript, Bootstrap, Google Gemini API

[GitHub repository](#)

[Visit NeurA](#)

ACHIEVEMENTS AND PARTICIPATIONS

- **SLIIT Codefest 2022-Finalists** : Won merit award at the hackathon under the tertiary category of SLIIT Codefest 2022 final round.
- **Openhack 2.0 Designathone 2023-Finalists - Organized by IIT** : Won merit award at the final round
- **HackX 2023 - Organized by University of Kelaniya** : Participations upto semi-finals
- **Code Rush 2023 - Intra Faculty Coding Competition** : Participation
- **Devthon 2024** : Participation
- **IDEALIZE 2024 - Organized by AIESEC in University of Moratuwa** : Participations upto second round
- **HackMoral 6.0 2024 Organized by INTECS** : 32 nd place in the batch 21 at faculty of IT (Individual)

EXTRA- CURRICULAR ACTIVITIES

- **FIT Moments** : Member of the HR pillar of IT Faculty Official Media Unit.
- **School cricket team (Hard-ball)** : Captain (2013 - 2015) , Member (2011 - 2016)
- **School Prefect** : 2010, 2016
- **Rotaract Club of UoM** : Organizing committee member - Binara Padura 2023
- **Mora Hiking Club** : Member of the hiking club

REFERENCES

Mr. Yashod Perera

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